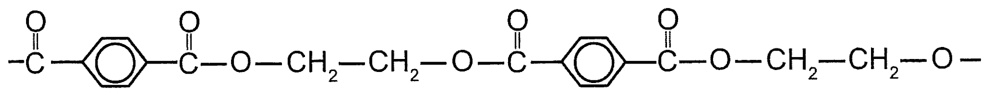
Organic Chemistry

Chapter
10

1. **[7 marks]** *5.2.13 (2015:32)*

Dacron is the trade name for a common polyester used in making clothes and water bottles. Part of its structural formula is given below:



(a) Draw the structural formula for the **two** monomers that react to form this polymer. [2]

Monomer one:

Monomer two:

(b) Name the other product of this polymerisation reaction. [1]

(c) Predict and explain the effect on the polyester's rigidity and melting point as the polymer chains increase in length. [4]

2. **[10 marks]** *5.2.1, 5.2.7, 5.2.3 (2015:34)*

Three different organic compounds were each tested with two reagents:
- acidified sodium permanganate solution and
- acidified propanoic acid.

Each organic compound has a molecular formula containing four carbon atoms, one oxygen atom and a number of hydrogen atoms.

The observations made are summarised in the following table.

Unknown organic compound	**Reagent added**	
	Acidified sodium permanganate solution	**Acidified propanoic acid**
1	no observable change	fruity smell
2	purple solution decolourises	no observable change
3	no observable change	no observable change

(a) Complete the table below, identifying the:
- functional group responsible for the observations made
- organic compound, by drawing its structural formula **or** giving its name. [6]

Unknown organic compound	**Functional group**	**Structural formula or name of the organic compound**
1		
2		
3		

(b) Draw the structural formula, showing all atoms of the organic product of the reactions of Compound 1 and Compound 2.

(i) Organic Compound 1 with the acidified propanoic acid. [2]

(ii) Organic Compound 2 with the acidified sodium permanganate solution. [2]

3. **[10 marks]** *5.2.14, 5.2.9 (2015:39)*

Amino acids are biologically-important organic compounds containing both amine (-NH_2) and carboxylic acid (-COOH) functional groups.

An important amino acid is 2-aminopropanoic acid; usually known as alanine. It is a component in more than a thousand different proteins, found in an array of foods and can be produced within the body.

Structural formula for alanine

$$\begin{array}{c} \quad\; NH_2 \\ \quad\; | \\ CH_3 - CH - COOH \end{array}$$

(a) Alanine is an alpha (α) amino acid. State the structural feature of alanine that allows it to be classified as an **alpha** (α) amino acid. [1]

(b) Use the following information to demonstrate that the molecular formula of alanine is the same as its empirical formula.

When 1.86 g of alanine was vaporised at 550.0 °C and 50.0 kPa pressure, it occupied a volume of 2.86 L. [4]

To consider the effect of having both an amine and a carboxylic acid functional group on the same molecule, amino acids can be compared with other organic compounds that have either:

- two amine functional groups on the same molecule (these compounds are called diamines)

or

- two carboxylic acid functional groups on the same molecule (these compounds are called dicarboxylic acids).

Amino acids have significantly higher melting points than diamines and dicarboxylic acids of similar mass and structure. This is illustrated in the table below.

Compound type	Example	Molar mass g mol^{-1}	Melting point °C
diamine	$CH_3-(CH_2)_4-\overset{\overset{NH_2}{\vert}}{CH}-NH_2$ hexane-1,1-diamine	116.2	39
dicarboxylic acid	$CH_3-\overset{\overset{COOH}{\vert}}{CH}-COOH$ methylpropanedioic acid	118.09	184
amino acid	$(CH_3)_2CH-\overset{\overset{NH_2}{\vert}}{CH}-COOH$ 2-amino-3-methylbutanoic acid (valine)	117.15	298

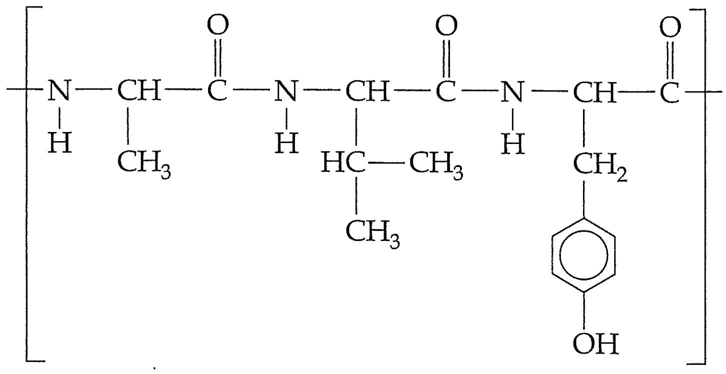
(c) Explain why amino acids form crystalline solids and have significantly higher melting points than other organic molecules of similar mass and structure. Refer to the information provided in the table above and include a labelled diagram using the amino acid, valine, to illustrate your answer. [5]

4. **[3 marks]** *5.2.8 (2016 SP:28b)*

Acetylsalicylic acid is a weak acid, and only partly ionises in water. It is poorly soluble in water, and far less soluble than a related compound, acetic acid (CH_3COOH). Explain why the water solubility of molecular acetylsalicylic acid is less than that of CH_3COOH.

5. **[6 marks]** *5.2.16, 5.2.14, 5.2.18 (2016 SP:31)*

Examine the structure below that represents a segment of the primary structure of insulin to answer the questions that follow.



(a) Circle **all** the peptide linkages (functional groups that link the monomers) represented in the above structure. *5.2.16* [1]

(b) Draw the molecular structures of the **three** α-amino acids that form this segment of insulin. *5.2.14* [3]

(c) The active form of insulin is made up of two polypeptide chains that contain five alpha helices. State the type of interactions that stabilises these secondary structures and the functional groups involved. *5.2.18* [2]

Type of interaction	Functional groups

5.2.1, 5.2.3 *(2016 SP:33)*

6. **[6 marks]**

Consider the following reactions and complete the tables that follow.

(a) An excess of butan-2-ol is oxidised by acidified $Na_2Cr_2O_7$ solution. *5.2.6* [3]

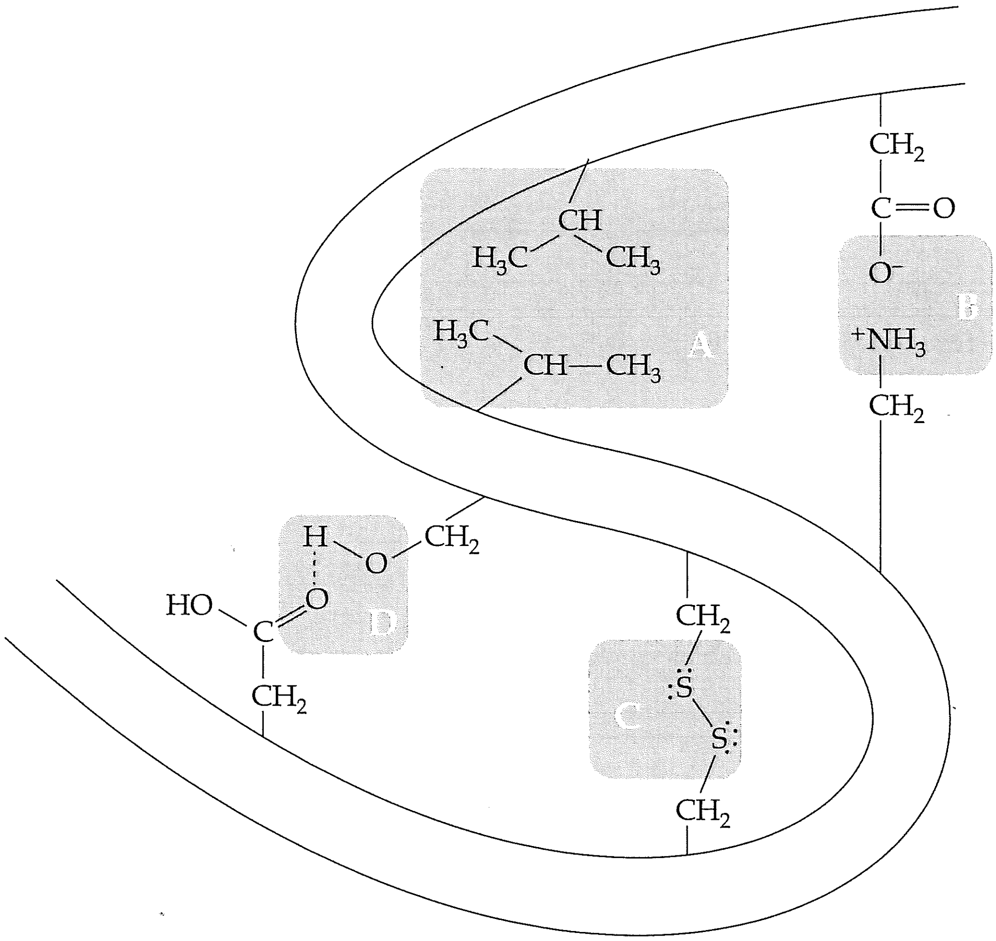
Observations	
Structural formula of organic product **(show all atoms)**	
Name of organic product	

(b) Butanoic acid reacts with methanol in the presence of concentrated H_2SO_4. *5.2.7* [3]

Observations	
Structural formula of organic product **(show all atoms)**	
Name of organic product	

7. **[5 marks]** *5.2.19 (2016 SP:36)*

The diagram below represents a segment of protein, showing the types of interactions that can occur between amino acid side chains to form the tertiary structure.



(a) Identify these types of interactions, labelled **A, B, C and D**, by completing the table below. [4]

Label	Type of interaction
A	
B	
C	
D	

(b) State what is meant by the 'tertiary structure' of a protein. [1]

5.2.1, 5.2.3 *(2016:27)*

8. **[9 marks]**

Write observations for the changes occurring when the substances below are mixed. In your answers include the appearance of the reactants and any product(s) that form.

(a) (i) methanol, pentanoic acid and sulfuric acid [2]

(ii) powdered magnesium carbonate and excess methanoic acid solution [2]

(iii) acidified potassium permanganate solution and excess propan-2-ol [2]

(b) Name the organic product and write the equation for the reaction when pentanal is added to a solution containing acidified sodium dichromate. [3]

Organic product: ____________________

9. **[9 marks]** *5.2.10, 5.2.11, 5.2.12, 5.2.18 (2016:29)*

Addition and condensation polymers are used in industry to produce a vast range of plastics.

Select **one** addition polymer you have studied and use it to complete parts (a) to (c).

(a) Draw and name the structure of the monomer used to produce this polymer. [2]

Name: ______________________________

(b) Draw and name the polymer, including at least **three** repeating units. [2]

Name: ______________________________

(c) State **one** use for this polymer, making reference to its relevant property/ies. [2]

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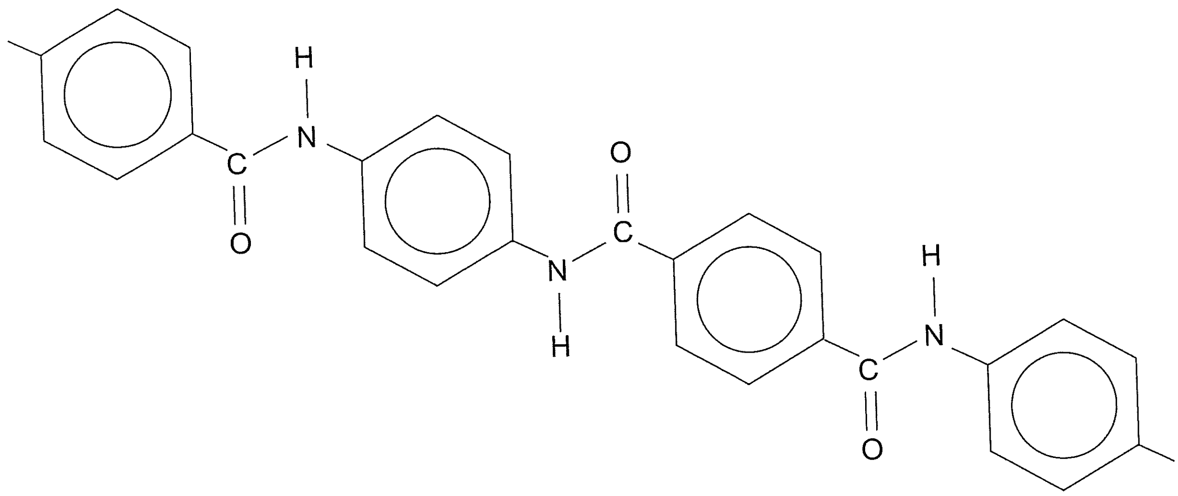
Kevlar is a condensation polymer utilised for its high strength. A section of the Kevlar polymer is drawn below.



(d) Draw the **two** monomers from which Kevlar is derived. [2]

Kevlar's high strength can be attributed in part to the hydrogen bonding that occurs between neighbouring chains. This is similar to a secondary structure of proteins.

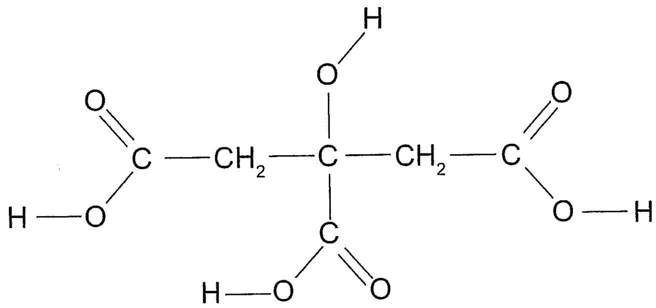
(e) To what secondary structure of proteins does this refer? [1]

10. **[5 marks]** *3.2.1, 5.2.1 (2016:33)*

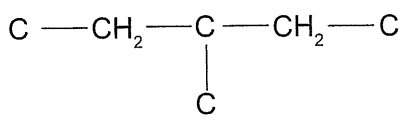
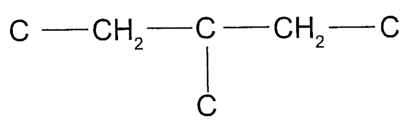
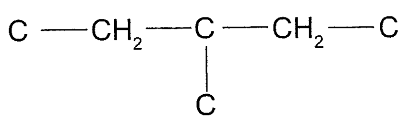
Citric acid, $C_6H_8O_7(aq)$, is a triprotic acid which reacts readily with solid sodium hydroxide, $NaOH(s)$.

(a) Write a balanced chemical equation for this reaction, showing all state symbols. [2]

The structure of $C_6H_8O_7$ is shown below.



(b) In the spaces below, complete the structures, showing **each** successive ionisation of the acidic hydrogen atoms. [3]

H^+ removed	Structure
First	
Second	
Third	

5.2.1, 5.2.8 (2016:35)

11. **[9 marks]**

For each of the three organic compounds identified in the table below:

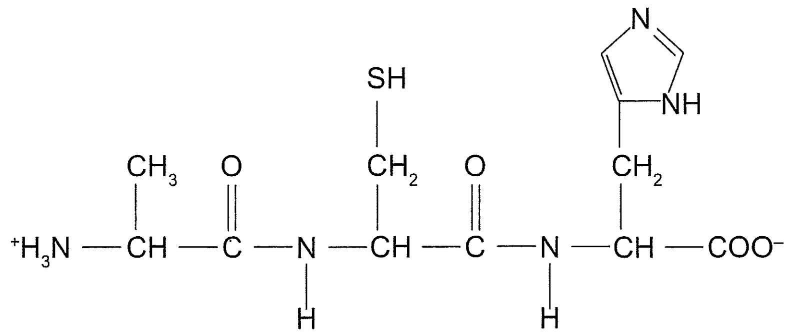
- use a structural formula to show the arrangement of **all** the atoms and **all** the bonds
- state **all** the intermolecular forces that exist between its molecules.

Organic compound	Full structural formula	All intermolecular forces
hexan-3-one		
1,1-difluoroethane		
butanamide		

12. **[11 marks]** *5.2.16, 5.2.13 (2016:36)*

Condensation reactions will take place between different α-amino acids and results in them being joined by peptide bonds. Structures produced by two α-amino acids are called dipeptides, while those produced by three are called tripeptides.

(a) Below is the structure of a particular tripeptide.



(i) Circle the peptide bonds on the structure. [2]

(ii) Name the **three** α-amino acids that reacted to form this tripeptide. [3]

One: ____________________

Two: ____________________

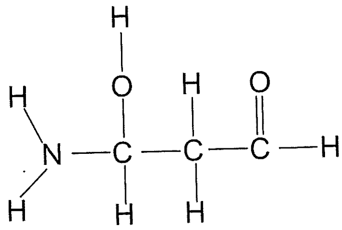
Three: ____________________

(b) Using the symbols (abbreviations) for these three α-amino acids, give **one** other polypeptide that can be formed from them. [1]

____

Alanine is one of the simplest examples of the twenty α-amino acids found in the human body. The structure below is an isomer of alanine.

(c) Circle and name each of the **three** functional groups on the isomer of alanine drawn below. [3]



(d) Draw a different isomer of alanine, showing clearly **all** atoms and **all** bonds. [2]

Isomer of alanine structure

13. **[5 marks]** *5.2.8 (2016:37)*

Pentane, pentanal and pentanoic acid all contain the same number of carbon atoms but display different physical properties. Their boiling points are given in the table below.

Organic compound	**Boiling point (°C)**
pentane	36.1
pentanal	102
pentanoic acid	186

Account for the difference in boiling points of the three compounds.

5.2.8 (2017:32)

14. **[9 marks]**

Ethanol and methanol are completely miscible (soluble) in water.

(a) By referring to any intermolecular forces present, describe the dissolving process as ethanol is added to water. [3]

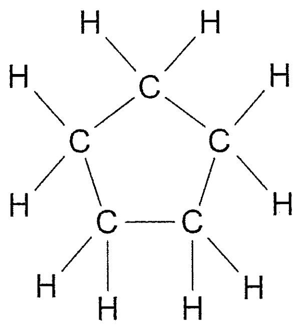
(b) Explain what happens to the solubility of alcohols in water as the hydrocarbon chain length increases. [3]

(c) For each of the following substances, list all force/s of attraction formed between the solute and solvent when each substance dissolves in water. [3]

Substance	**Force/s of attraction with water**
Propanal	
Methanoic acid	
Sodium chloride	

15. **[5 marks]** *5.2.5 (2017:35)*

There are a number of different isomers with the molecular formula of C_5H_{10}. These include chain isomers and cyclic isomers such as cyclopentane, which is shown here.



(a) Draw one chain isomer for C_5H_{10} that satisfies each of the following types. For each isomer, show all atoms and all bonds. [2]

Type	**Diagram**
trans isomer	
cis isomer	

Chemical tests (adding reagent/s) can be used to distinguish between chain and cyclic isomers in this question.

(b) In the table below suggest a distinguishing test by stating the reagent/s used and the observations expected for any reaction with each isomer. [3]

Reagent/s		
	Cis/trans chain isomer	**Cyclic isomer**
Observations		

5.2.18 (2017:40)

16. **[15 marks]**

The properties of human hair can be attributed to it being composed almost entirely of the strong fibrous protein, keratin.

Structure of keratin:
- Keratin is a polypeptide and consists of a repeating pattern of amino acids.
- Common amino acids in keratin, in order from most to least abundant, are: cysteine (17.5%), serine, glutamic acid, threonine, glycine, leucine, valine, arginine, aspartic acid and alanine (4.8%).

(a) Draw a section of the polypeptide that is composed of the three most abundant amino acids found in keratin. [4]

(b) With reference to the structure drawn in part (a), state three types of attractive forces/ bonding other than dispersion forces, that can occur between neighbouring keratin polypeptide chains. [3]

One: ______

Two: ______

Three: ______

(c) Describe the α-helix structure of keratin. [2]

______

______

______

______

One of the physical properties of hair is its capacity to absorb water, increasing a strand's diameter by roughly 20%.

(d) State why hair can absorb water. [1]

______

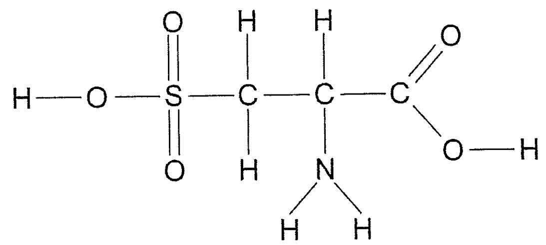
______

Keratin is often chemically analysed for cysteine, due to its effect on the strength of hair. One method of determining the proportion of cysteine is titration with bromide in an acidic solution. Under these conditions, the cysteine is oxidised to cystine and then to cysteic acid.

Two cysteine molecules joined together by a disulfide bond is called cystine.

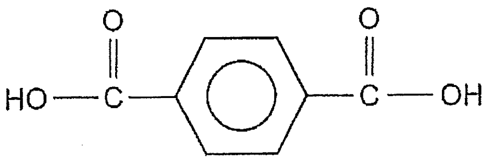
(e) Draw the structure of cystine. [2]

(f) On the structural formula of cysteic acid drawn below, circle and label any functional groups as acidic or basic. [3]

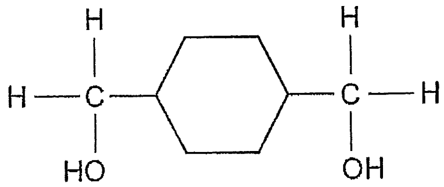


17. **[7 marks]** *5.2.13 (2018:30)*

Polycyclohexanedimethyl terephthalate glycol, (PCTG), is a strong, chemically-resistant polymer that is food-safe. The monomers needed to synthesise PCTG are terephthalic acid and 1,4-cyclohexanedimethanol, as shown below.



terephthalic acid



1,4-cyclohexanedimethanol

(a) In the space below, draw the structural formula of PCTG, showing **two** repeating units. [2]

(b) State the name or give the formula of the by-product of this polymerisation process. [1]

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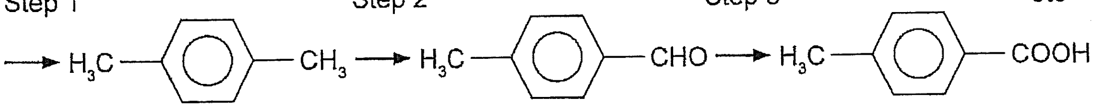
The following flow diagram shows some of the steps needed to synthesise terephthalic acid.



(c) Name **two** reagents that could be used to synthesise (C) from (B) in Step 3. [2]

One: ____________________________________

Two: ____________________________________

(d) Write a balanced half-equation to show (B) reaction to form (C). [2]

18. **[11 marks]** *1.1, 5.2.8 (2018:33)*

Consider the compounds and their properties listed in the table below.

Compound	Boiling point (°C)	Solubility in water (g L^{-1})
Butane C_4H_{10}	-0.5	0.061
Butan-1-ol $C_4H_{10}O$	117	73.0
Butanone C_4H_8O	79.6	27.5

(a) Given that the molecular formulas indicate that the compounds contain the same number of carbon atoms and differ only in the number of one or two hydrogen or oxygen atoms, propose an hypothesis for why there is a variation in the boiling points of these compounds. [2]

(b) Explain why these organic compounds have very different solubilities in water. [6]

Butanoic acid, $C_4H_8O_2$, is another organic compound that contains four carbon atoms in each molecule and, like butan-1-ol, it is a colourless liquid.

(c) Complete the table below to describe a chemical test that could be used to distinguish between butan-1-ol and butanoic acid by stating the reagent/s used and the distinguishing observations. [3]

Reagent/s used		
Substance being tested	**Butan-1-ol**	**Butanoic acid**
Observations		

19. **[6 marks]** *5.2.6 (2018:34)*

For the molecular formula $C_6H_{12}O$ draw **two** different structural isomers, one which can be readily oxidised by acidified dichromate solution and one which cannot be readily oxidised by acidified dichromate solution. Show all atoms.

Isomer that **can** be readily oxidised by acidified dichromate solution.

Isomer that **cannot** be readily oxidised by acidified dichromate solution.

20. **[14 marks]** *5.2.16, 5.2.15 (2018:39)*

The Atlantic Iongfin inshore squid is able to blend into its surroundings and seemingly disappear. It does this by reflecting light using specialised cells. The squid tunes and adapts the reflection of light from these cells by using a class of proteins called reflectins.

The amino acid sequences of some reflectins from this squid have been characterised. A small sequence from one of the reflectins is shown below.

– Tyr – Val – Asn –

(a) Draw the full structural formula of this section of the reflectin. Show all hydrogen atoms. [3]

(b) Circle **one** peptide bond in the structure that you drew in part (a). [1]

The amino acid leucine is also found in reflectin.

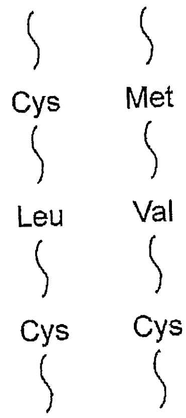
(c) Draw the full structural formula of leucine, Leu, in each of the conditions specified below. Show all hydrogen atoms. [4]

Low pH (acidic)

High pH (basic)

(d) Explain why the structure of Leu is pH dependent. [3]

Consider the following amino acids found on neighbouring protein chains as they come into proximity to each other.



(e) Identify the pair **most** strongly attracted to each other. Justify your choice. [3]

21. **[12 marks]** *5.2.1, 5.2.2, 5.2.3 (2019:33)*

Organic molecules have a hydrocarbon skeleton and can contain functional groups that are responsible for the molecules' characteristic chemical properties.

Complete the following tables by

(i) writing the structural formula of each compound listed

(ii) writing the structural formula of the organic product from the reaction

(iii) naming the organic product from the reaction.

When writing the structural formula, show the bonds between carbon atoms and within any functional group e.g.

$$\begin{array}{c} CH_3 - CH_2 - \underset{\displaystyle O}{\underset{\|}{C}} - CH_3 \end{array}$$

<table>
<tr><th colspan="2">Name of compound</th><th>Structural formula of compound</th></tr>
<tr><td colspan="2">pent-2-ene</td><td></td></tr>
<tr><td rowspan="2">Reacts with $Br_2(aq)$</td><td>Structural formula of organic product</td><td></td></tr>
<tr><td>Name of organic product</td><td></td></tr>
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<table>
<tr><th colspan="2">Name of compound</th><th>Structural formula of compound</th></tr>
<tr><td colspan="2">ethanal</td><td></td></tr>
<tr><td rowspan="2">Reacts with $KMnO_4(aq)$/ $H^+(aq)$</td><td>Structural formula of organic product</td><td></td></tr>
<tr><td>Name of organic product</td><td></td></tr>
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<table>
<tr><th colspan="2">Name of compound</th><th>Structural formula of compound</th></tr>
<tr><td colspan="2">butanoic acid</td><td></td></tr>
<tr><td rowspan="2">Reacts with $Na_2CO_3(aq)$</td><td>Structural formula of organic product</td><td></td></tr>
<tr><td>Name of organic product</td><td></td></tr>
</table>

22. [18 marks] *5.2.12, 5.2.13 (2019:38)*

Polymethyl methacrylate and polycarbonate are two polymers that are used as alternatives to glass. Polymethyl methacrylate is more commonly known as Perspex or plexiglass and is an addition polymer, while polycarbonate is a type of condensation polymer.

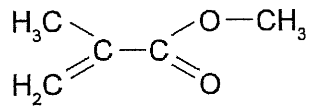
Both polymers are transparent to visible light and have other properties as listed below.

Polymethyl methacrylate	**Polycarbonate**
lightweight	moderate chemical resistance
moderate UV resistance	high heat resistance
low impact strength	high impact strength
low chemical resistance	low scratch resistance
low heat resistance	low UV resistance

(a) For the following uses as an alternative to glass, identify which polymer would be the more appropriate. Justify your choice of polymer by comparing the effect of **two** relevant properties as listed for both polymers. [4]

Use	**Choice of polymer**	**Justification**
Skylight		
Safety glasses		

The monomer, methyl methacrylate, can be formed from the esterification of methanol and methacrylic acid (2-methylprop-2-enoic acid). The structural formula of methyl methacrylate is shown below.



(b) Write a balanced equation for the esterification of methanol and methacrylic acid. Show the full structural formula of each species in the equation. [4]

Methyl methacrylate can undergo addition polymerisation to form polymethyl methacrylate.

(c) Draw a section of a polymethyl methacrylate showing **all** atoms and at least **three** repeating units of the monomer. [3]

One method for the production of methacrylic acid is by the following oxidation.

$$\underset{\text{methylpropenol isomer}}{C_4H_8O} \xrightarrow{\text{oxidation}} \underset{\text{methacrylic acid}}{C_4H_6O_2}$$

(d) Suggest an assumption that **must** be made regarding the mole ratios of product to reactant for this reaction and then determine the mass of the methylpropenol isomer required to produce 1.50 tonne of methacrylic acid if the efficiency of this oxidation is 65%. (Note: 1 tonne = 1000 kg.) [5]

Assumption: ____

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Calculation: ____

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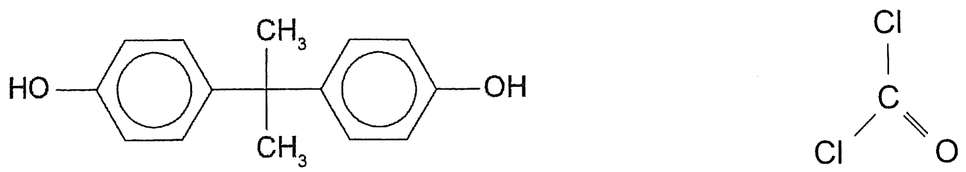
Polycarbonates are condensation-type polymers for which the by-product is hydrogen chloride instead of water.

The two monomers for polycarbonate are shown below.



(e) Why is polymethyl methacrylate classified as an addition polymer, while polycarbonate is classified as a condensation polymer? [2]

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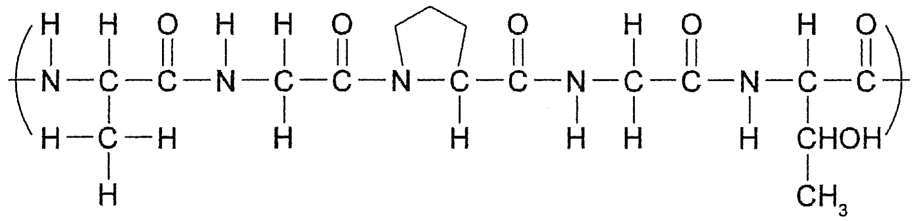
23. **[12 marks]** *5.2.17, 5.2.15, 5.2.18 (2019:41)*

When insects touch a spider's web they become stuck and therefore, easy prey for the spider. The insects become stuck because the web is coated with a glue-like substance produced by the spider. The 'spider glue' consists of water, proteins, ionic salts and polar carbon compounds.

The structural formula given below shows a small section of a spider glue protein.



(a) List the names of the amino acids in the order in which they were drawn in the section of the protein given above. Do **not** use abbreviations. [3]

(b) Circle **one** peptide bond in the above structure. [1]

(c) What is the difference between the primary structure and the secondary structure of a protein? [2]

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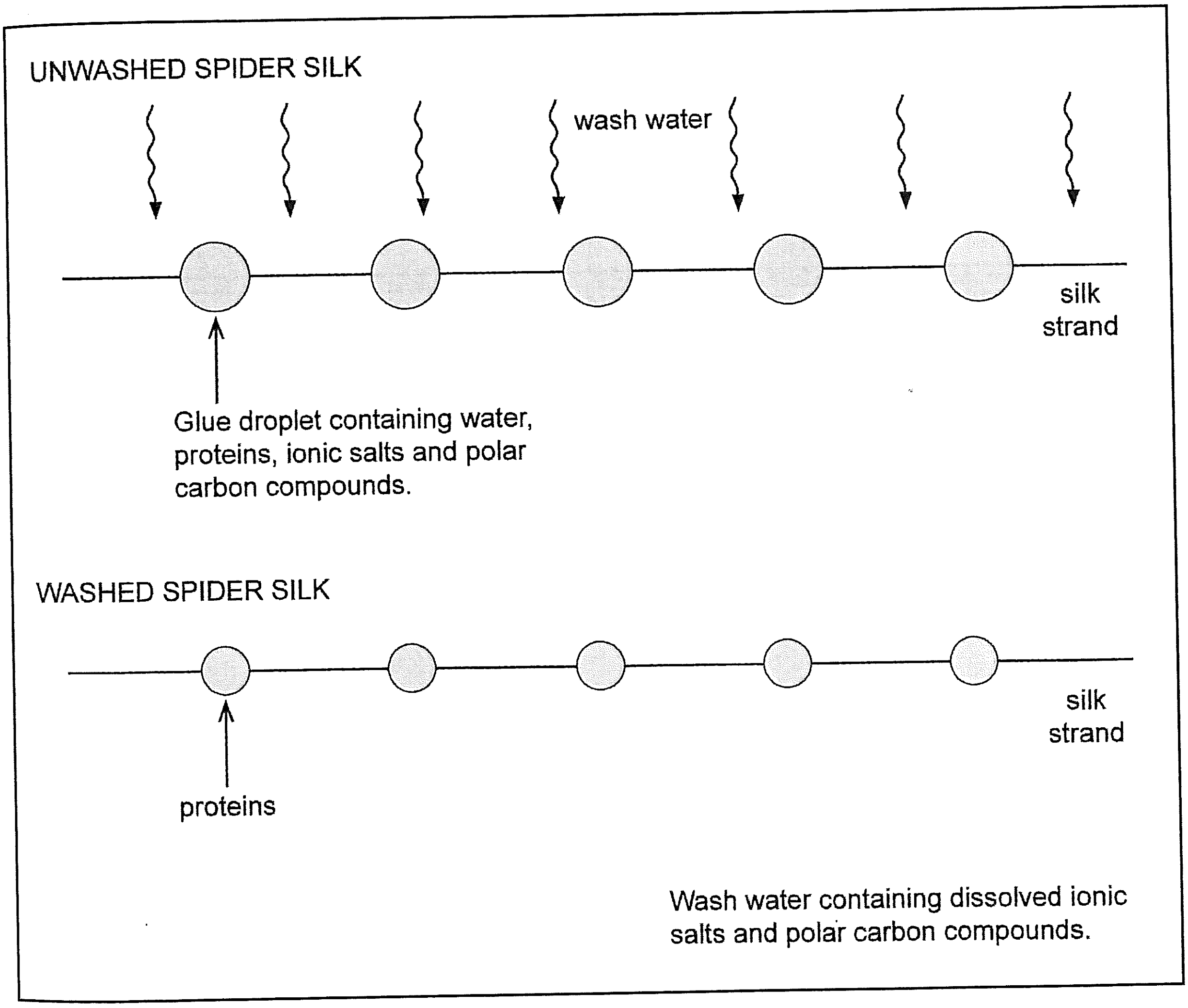
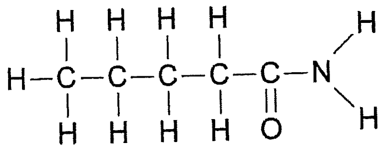
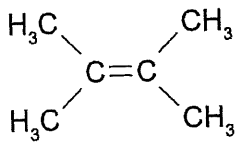
When spider glue is washed with water, the ionic salts and polar carbon compounds dissolve. The proteins do not dissolve and remain on the silk strand. The following diagram shows what happens.



(d) Explain why the polar carbon compounds dissolve in water but the proteins do not. Illustrate your answer with the aid of a labelled diagram. [6]

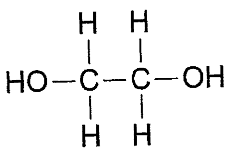
24. **[4 marks]** *5.2.4 (2020:26)*

Complete this table by giving the IUPAC name or full structural formula of the indicated organic compounds. All hydrogen atoms must be shown.

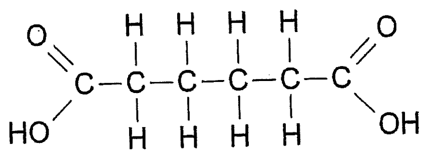
Full structural formula	IUPAC name
	
	
	heptan-2-amine
	hexan-3-one

25. **[5 marks]** *5.2.10, 5.2.12 (2020:28)*

Poly(ethylene adipate) is an inexpensive, biodegradable polymer. It is formed when ethylene glycol and adipic acid react. The structural formulae of these two monomers are shown below.



ethylene glycol



adipic acid

(a) Draw the structural formula of poly(ethylene adipate). Show two repeating units. [2]

(b) Classify poly(ethylene adipate) according to the:

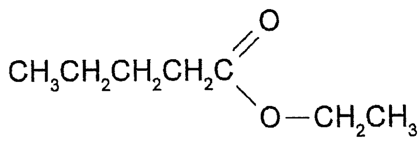
(i) functional group or groups present in its structure. [1]

(ii) type of reaction resulting in its formation. [1]

(c) Identify a different type of reaction that results in the formation of a polymer. [1]

26. **[9 marks]** *5.2.7 (2020:33)*

A chemist wanted to add a fruity fragrance to an air freshener that he was developing. A colleague suggested the compound ethyl pentanoate which has an apple-like fragrance. The structure for ethyl pentanoate is shown below.



The chemist wanted to check the fragrance of this compound to make sure that it was suitable but there was no ethyl pentanoate in the chemist's laboratory. The only organic substances that the chemist had were a:

- commercial gas cylinder containing ethene
- bottle of pentan-2-one
- bottle of pentan-1-ol
- bottle of pentanal.

Ethyl pentanoate can be synthesised from one or more of the organic substances in the above list in **three** steps.

Describe the steps that will allow the chemist to synthesise ethyl pentanoate. Include balanced equations for all reactions that occur, using molecular formulae for organic compounds. Any inorganic compounds deemed necessary can be used in the procedure. It is not necessary to specify how the products of a particular reaction will be isolated before use in another reaction.

Step One:

Step Two: ____

____

____

____

____

____

____

____

Step Three: ____

____

____

____

____

____

____

____

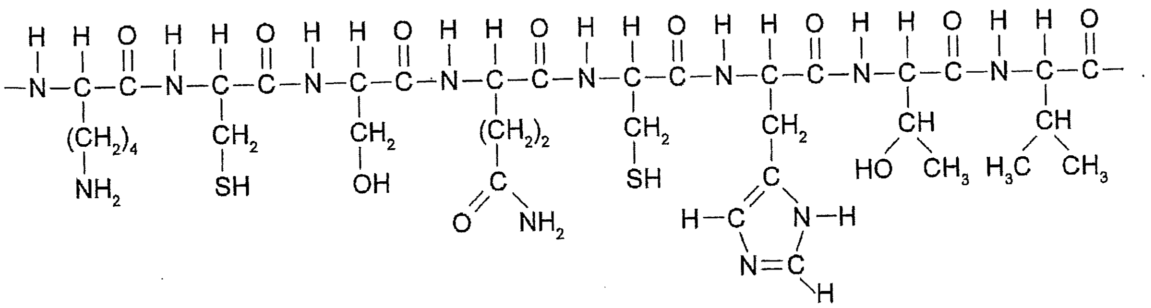
27. **[11 marks]** *5.2.16, 5.2.17, 5.2.18, 5.2.19 (2020:35)*

Cytochrome C is a protein found in the cells of many organisms. A biochemist analysed the Cytochrome C from a human and a grey whale to establish their respective α-amino acid sequences.

(a) What protein structure level does the α-amino acid sequence represent? [1]

____

The structural formula of a small segment of human Cytochrome C, as written by the biochemist in her notebook, is shown below.



The biochemist wrote the sequence of α-amino acids in the corresponding grey whale Cytochrome C segment in an abbreviated form

– Lys – Cys – Ala – G1n – Cys – His – Thr – Val –

(b) Identify **one** similarity and **one** difference between the given α–amino acid sequences of human and grey whale Cytochrome C. [2]

Similarity: ____

____

____

Difference: ____

____

____

The biochemist examined the overall three–dimensional folded shape of grey whale Cytochrome C. The biochemist did this by identifying the predominant types of interactions occurring between the side chains of α–amino acids located near each other in grey whale Cytochrome C. Three of the α–amino acid pairs considered by the biochemist are shown in the following table.

(c) Complete the following table by identifying the predominant side chain interaction for each α–amino acid pair. [3]

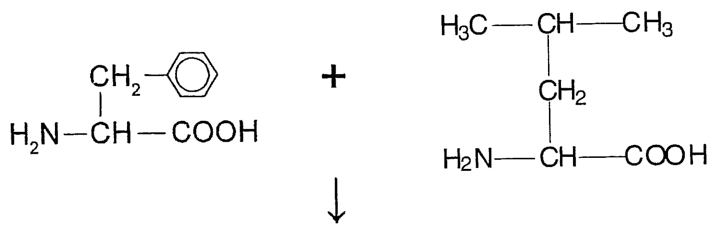
α–amino acid pairs	**Predominant side chain interaction**
Ala and Val	
G1n and His	
Cys and Cys	

(d) The biochemist found that both human and grey whale Cytochrome C contain several alpha helices but no beta-pleated sheets. What protein structure level do alpha helices and beta-pleated sheets represent? [1]

____

Further analysis of human Cytochrome C showed that there was a segment where two other α–amino acids (phenylalanine and leucine) were adjacent to each other. The biochemist obtained pure samples of each of these amino acids and set up an experiment to facilitate their reaction with each other.

(e) Write a balanced equation, using condensed structural formulae, for a reaction that occurs between phenylalanine and leucine. [2]



(f) The biochemist decided to examine how the structure of leucine changes with solution pH. Complete the following table by drawing the structural formula of leucine at the indicated pH. [2]

Structural formula of leucine	**pH**
	acidic
	basic

Potassium chloride solution and bromine water

Equation 'no reaction' – bromine cannot spontaneously displace chlorine.
Observation(s) 'no visible reaction' – an orange solution was added to a colourless solution and the mixture remained orange.

13.(2020:36) a) To be a functioning galvanic cell it must be spontaneous – the E° must be positive with 1 mol L^{-1} solutions. So magnesium and $Cu^{2+}(aq)$ meet this criterion. The cathode must not react with the $Cu^{2+}(aq)$ so it must be copper or graphite. The salt bridge should be ionic and soluble but not form precipitates so a carbonate would not be a good choice. (i) magnesium (ii) copper or graphite (iii) (1.0 mol L^{-1}) magnesium sulfate (solution) (iv) (1.0 mol L^{-1}) copper(II) sulfate (solution)

b) In the external wire an arrow from left to right: →

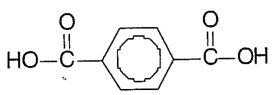
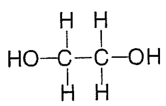
c) Oxidation occurs at the anode. Reduction occurs at the cathode.
Anode half-equation – $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$
Cathode half-equation – $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

d) Note the question asks for the units to be included

$Mg(s) \rightleftharpoons Mg^{2+}(aq) + 2e^-$ E° = +2.36V
$\underline{Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)}$ $\underline{E° = +0.34V}$
Total +2.70 V

e) i) A salt bridge provides ions to each cell to maintain neutrality. Sometimes this is expressed as completing the circuit.
ii) In this cell, at the anode, positive $Mg^{2+}(aq)$ ions are being formed (oxidation) while at the cathode, $Cu^{2+}(aq)$ ions are being destroyed (reduction).
Negative ions move through the salt bridge to return the charge to zero at the anode half-cell where $Mg^{2+}(aq)$ ions are being formed.
Positive ions move through the salt bridge to return the charge to zero at the cathode half-cell where $Cu^{2+}(aq)$ ions are being destroyed and the solution would have been left with a negative charge.

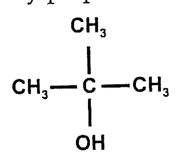
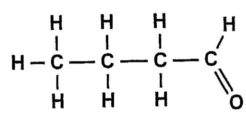
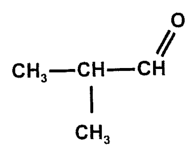
Chapter 10: Organic Chemistry

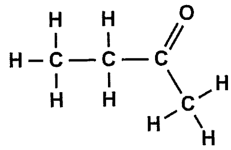
1.(2015:32) a)  and 

b) Water

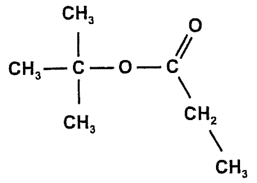
c) Increasing the length of the polymer chain increases the dispersion forces of the polymers. The number of =O dipoles increases so the number of dipole attractions along the length of the polymer chains increases.
- This increases the magnitude of its interactions with neighbouring chains making the polymer more rigid
- More energy is required to overcome the attraction of the chains of the polymer from each other thereby raising the melting point

2.(2015:34) a)

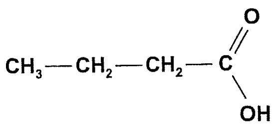
Unknown organic compound	Functional group	Structural formula or name of the organic compound
1	alcohol	methylpropan-2-ol 
2	aldehyde	butanal or methylpropanal  

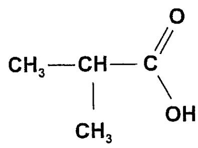
3	ketone	Butanone 

b) i)



ii) either of





3.(2015:39) a) The $-NH_2$ group and the $-COOH$ group must be attached to the same (α) carbon atom

b) From the given formula, the empirical formula is $C_3NH_7O_2$

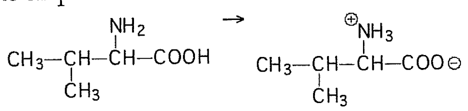
Empirical formula mass ($C_3NH_7O_2$) = 89.094

$n_{(alanine)}$ = PV/RT = (50.0 × 2.86) / (8.314 × 823.15) = 0.020895 mol

$M_{(alanine)}$ = m/n = 1.86 / 0.02089 = 89.02 g mol^{-1}

The empirical formula mass is the same as the calculated molecular formula mass

c)



This transfer produces a zwitterion. It has a positive and a negative ionic charge. It has no overall electrical charge. This strong ionic attraction between ions forms a crystalline lattice in the solid state. The di-amine and di-carboxylic acid bond using weaker hydrogen bonding. These ionic attractions require more energy to break and so the zwitterions have high melting points.

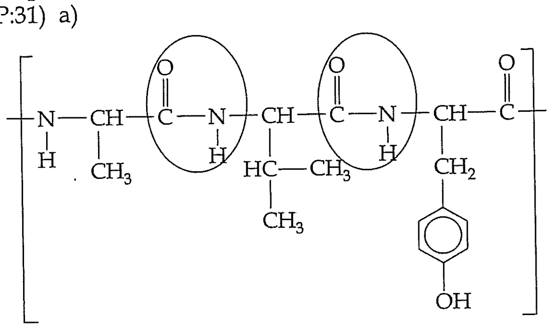
4.(2016 SP:28b) To dissolve well the bonds between solutes must be broken and the bonds between solvents must be broken and the new solute solvent bond must form.

Acetylsalicylic acid bonds with itself with strong dispersion forces (due to the large molecular size). Acetic acid molecules engage with each other with weak dispersion forces (small molecule), hydrogen bonding and dipole/dipole forces. Water engages with itself primarily with hydrogen bonding.

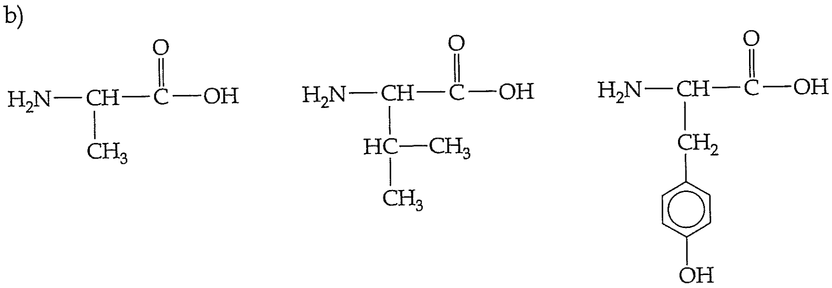
Acetylsalicylic acid and water molecules and water cannot overcome the large dispersion forces of acetylsalicylic acid and interact poorly thus are immiscible.

Acetic acid forms hydrogen bonds with water using the OH group and the carbonyl group as well as weak dispersion forces and so dissolves well.

5.(2016 SP:31) a)



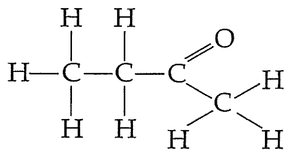
b)



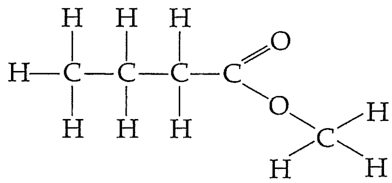
c)

Type of interaction	Functional groups
hydrogen bonding	carbonyl and amide groups

6.(2016 SP:33) a)

Observations	the solution turns from orange to green
Structural formula of organic product	
Name of organic product	butanone

b) Butanoic acid reacts with methanol in the presence of H_2SO_4 solution.

Observations	a fruity smell develops (and a new layer forms above the reactants)
Structural formula of organic product (show all atoms)	
Name of organic product	methylbutanoate

7.(2016 SP:36) a)

Label	Type of interaction	Label	Type of interaction
A	dispersion forces	**C**	covalent bond or disulfide bond
B	ionic bond	**D**	hydrogen bond

b) The 'tertiary structure' is the overall 3-dimensional shape

8.(2016:27) a)

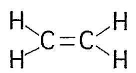
i)
- three colourless liquids mix
- an immiscible layer appears on the surface
- a fruity smell produced

ii)
- Note the acid is inxs – this means there will be no $MgCO_3$ left over
- a white solid is added to colourless liquid
- a colourless odourless gas is produced
- the white powder dissolves and a colourless solution is formed

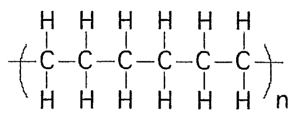
iii)
- Note the alcohol is inxs which means all the MnO_4^- will react
- a purple liquid is mixed with a colourless liquid
- the purple colour fades to very pale pink (almost colourless)
- a new paint smell can be detected

b)
- Pentanoic acid
- $Cr_2O_7^{2-} + 3\ C_5H_{10}O + 8\ H^+ \rightarrow 2\ Cr^{3+} + 3\ C_5H_{10}O_2 + 4\ H_2O$

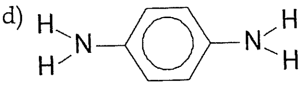
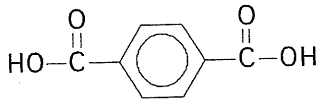
9.(2016:29) a) As it is an addition polymer it must have a double bond. The simplest and easiest answer is ethene and polyethene but you could use PVC, styrene, propene, Teflon etc.
Example.

Ethene	

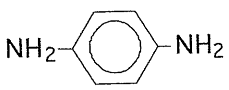
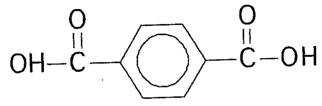
b) Note that the polymer has bonds on each end.

Polyethene	

c) This answer is taken from the syllabus. Note you need to say what it is used for and why.
If you used polyethene you could say cling-wrap, or take-away containers BECAUSE it is strong, transparent, flexible and cheap.

d)  

Note these structures below would be wrong if we wrote them. Can you see why?

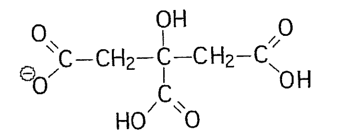
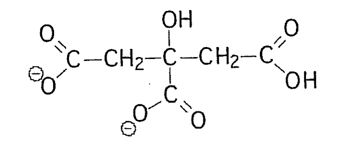
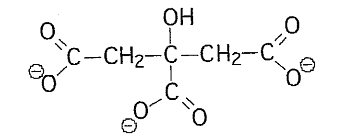
e) β-pleated sheets

10.(2016:33) a) $C_6H_8O_7(aq) + 3\ NaOH(s) \rightarrow Na_3C_6H_5O_7(aq) + 3\ H_2O(\ell)$

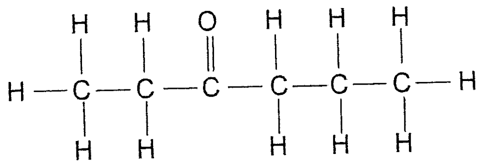
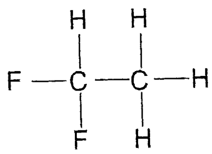
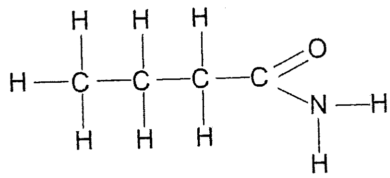
or

$C_6H_8O_7(aq) + 3\ NaOH(s) \rightarrow 3\ Na^+(aq) + C_6H_5O_7^{3-}(aq) + 3\ H_2O(\ell)$

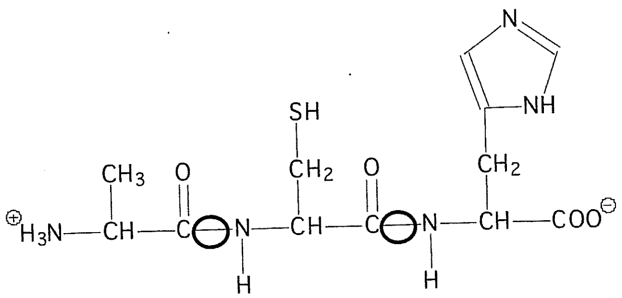
b)

		
Note the alcohol group is not considered acidic in this situation		

11.(2016:35)

Organic compound	Full structural formula	All intermolecular forces
hexan-3-one		dispersion dipole-dipole
1,1-difluoroethane		dispersion dipole-dipole
butanamide		dispersion hydrogen bonding dipole-dipole

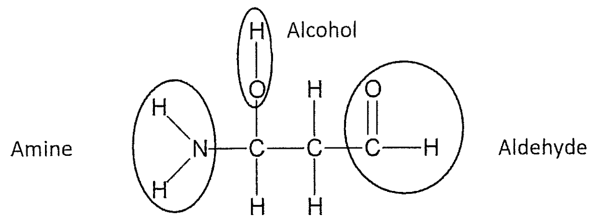
12.(2016:36) a) i)



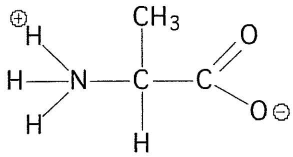
ii) Alanine, Cysteine, Histidine

b) This question requires you to understand amino acids can polymerise in any order. The answer must be different from the example given Ala-Cys-His, so Ala-His-Cys or His-Ala-Cys or any other combination is fine.

c)



d)



Any isomer is acceptable including the zwitterion. Be careful not to draw alanine on the isomer from part c)

13.(2016:37) To boil a molecular substance such as these three examples we must overcome the total secondary forces between the molecules.

Pentane, pentanal and pentanoic acid all exhibit dispersion forces due to the presence of electrons. As they contain the same number of carbon atoms the dispersion forces will be about equal.

Both pentanal and pentanoic acid have a carbonyl group that has a polar C=O bond. This gives these molecules dipole-dipole forces between them. Thus they have a higher boiling point than pentane. (note pentanal can accept hydrogen bonds from water but in this question there aren't any so it's irrelevant)

Only pentanoic acid has an –O–H configuration that allows the molecules to hydrogen bond to each other. Thus as it has dispersion forces, dipole-dipole bonds and hydrogen bonds it has the highest boiling point. Please note that the desk you are writing on probably has a plastic top made of long carbon chains held together by dispersion forces. It is very strong. It is not correct to say substances with hydrogen bonds have higher boiling points than substances with dispersion forces so pentanoic acid has a higher BP. It is the sum of all three that counts.

14.(2017:32) a) This is a typical question asked about solubility. It demands you talk about bond breaking in the solvent and the solute and bond forming in the solution. You need to discuss breaking and forming all the types of bonds including hydrogen bonds, dipole/dipole and dispersion forces. You should offer evidence of disruption of existing forces and formation of new forces.

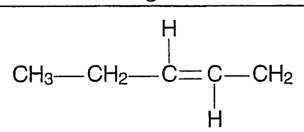
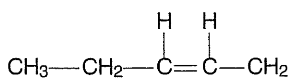
Ethanol interacts with ethanol by dispersion forces and hydrogen bonding. Water interacts with water molecules by hydrogen bonding and dispersion forces. The strength of the interaction between ethanol molecules and water molecules are stronger than the internal interactions of both so ethanol dissolves in water – therefore they are miscible

b) The solubility of alcohols in water decreases as the hydrocarbon chain length increases. This is due to the decreasing dominance of the hydrogen bond and the growing influence of the dispersion forces. As in the previous question the discussion turns on the idea that the difference between the energy released in the formation of new forces of attraction and the energy required to overcome the existing forces of attraction increases and so alcohols become more dispersion force predominant and less hydrogen bond dominant so they are less attractive to water as water interacts very poorly with dispersion forces so the solubility decreases.

c) The question asked for <u>all</u> secondary forces

Propanal	Hydrogen Bonding, dispersion forces, dipole-dipole forces
Methanoic acid	Hydrogen Bonding, dipole-dipole, dispersion forces, and, when ionized,– ion-dipole
Sodium chloride	Ion-dipole forces, dispersion forces

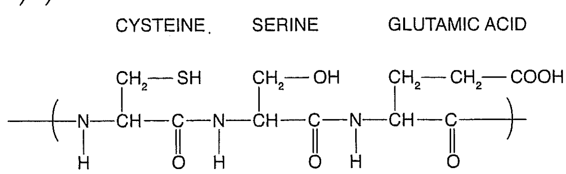
15.(2017:35) a) A chain isomer means we shift carbons to change the chain length.

Type	Diagram
trans isomer	
cis isomer	

b)

Reagent/s	Br_2 *(aq)*	
	Cis/trans chain isomer	**Cyclic isomer**
Observations	The orange bromine solution will quickly decolourise	The orange colour will remain.

16.(2017:40) a)

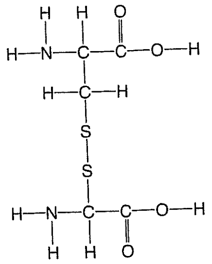


b) Disulfide bridges, Hydrogen bonding, Dipole-dipole interactions

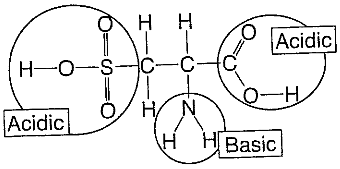
c) This is a secondary bond. It is a hydrogen bond between two amide groups in the primary structure. The hydrogen is between the N–H and the O in the C= O bond and causes a spiral to form.

d) Water can donate and accept hydrogen bonds with the amide group in hair.

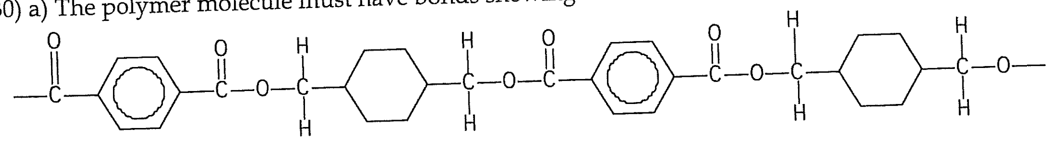
e)



f)

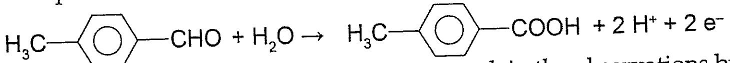


17.(2018:30) a) The polymer molecule must have bonds showing at each end.



b) Water – it was a condensation reaction.

c) To get these marks you must say <u>acidified</u> potassium dichromate <u>solution</u> and <u>acidified</u> potassium permanganate <u>solution</u> (or sodium). Unless its acidified it won't work and if you don't say 'solution' then we assume powder and that will not work either.

d) 

18.(2018:33) a) Note that this question did not ask you explain the observations but instead is examining your understanding of the term 'hypothesis'. You must provide an experimentally testable idea such as 'The boiling point of similar molar mass substances increases more by adding an oxygen atom than by removing two hydrogens'

b) To clarify your thinking a question like this might best be answered under headings.

<u>Water</u>
In this case water is the solvent and the water molecules are bound by solvent – solvent interactions called hydrogen bonds. Each water can accept two hydrogen bonds from two other water molecules and it can donate two more to other water molecules – a total of four hydrogen bonds. If a solute is to dissolve in water the solvent-solvent bonds and the solute–solute bonds must be broken when the solvent–solute bonds form and enough energy must be released to break the solvent–solvent bonds and the solute-solute bonds.

<u>Butane</u>
Butane is a non-polar molecule incapable of engaging in dipole–dipole or hydrogen bonding. It bonds to other butane molecules with dispersion forces and insufficient energy is released when interacting with water to break the solvent–solvent hydrogen bonds and the solute–solute dispersion forces so it does not dissolve.

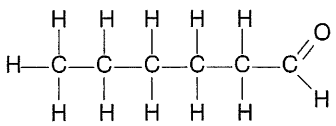
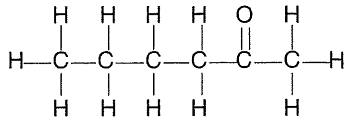
<u>Butanone</u>
Butanone is capable of dispersion force interactions, dipole–dipole interactions with other butanone molecules and can engage in hydrogen bonding with water by accepting two hydrogen bonds with its lone pairs on its C=O group. It must break its dipole-dipole interaction and break the solvent–solvent hydrogen bonds to dissolve and evidently this is energetically favourable.

<u>Butan-1-ol</u>
Butanol can engage in dispersion forces and both donate and accept hydrogen bonds. Each butanol can accept two hydrogen bonds from two other water molecules and it can donate one more to other alcohol or water molecules – a total of three hydrogen bonds. It demonstrates this attraction with its high boiling point. Water too has the capacity to donate two hydrogen bonds and accept two hydrogen bonds. Thus, when mixed with water, the solute–solvent success shows that the hydrogen bonds are able to release enough energy to overcome the solvent–solvent hydrogen bonds and solute-solute hydrogen bonds making it quite soluble in water.

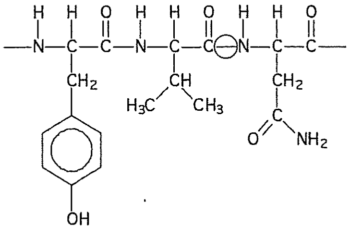
c)

Reagent/s used	Acidified potassium permanganate solution	
Substance being tested	Butan-1-ol	Butanoic acid
Observations	Purple solution decolourises	Purple solution remains purple – no observable change

19.(2018:34)

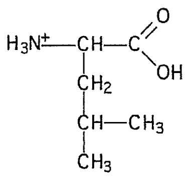
Isomer that can be readily oxidised by acidified dichromate solution.	Isomer that cannot be readily oxidised by acidified dichromate solution.
 An aldehyde	 A ketone

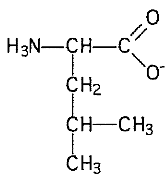
20.(2018:39) a)



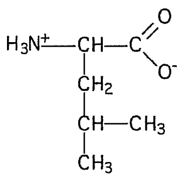
b) circled above

c) Low pH High pH





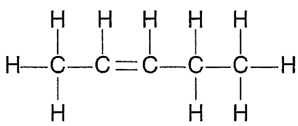
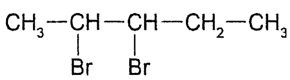
d) At neutral pH the amino acid exists as a zwitterion.

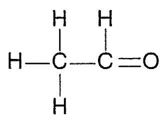
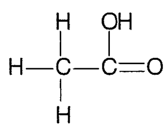


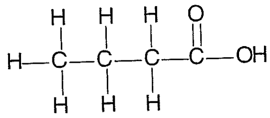
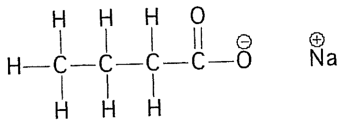
At low pH and high pH the structure changes as shown at left, acting as a base and as an acid.

e) Cys and Cys will establish a disulfide bridge between them. This is a covalent bond that is much stronger than hydrogen bonding or dispersion forces between other amino acids.

21.(2019:33)

Name of compound		Structural formula of compound
Pent-2-ene		
Reacts with $Br_2(aq)$	Structural formula of organic product	
	Name of organic product	2,3-dibromopentane

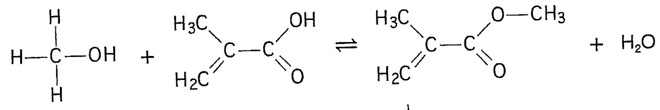
Name of compound		Structural formula of compound
Ethanal		
Reacts with $KMnO_4(aq)/H^+(aq)$	Structural formula of organic product	
	Name of organic product	Acetic Acid or Ethanoic Acid

Name of compound		Structural formula of compound
Butanoic acid(*aq*)		
Reacts with $Na_2CO_3(aq)$	Structural formula of organic product	
	Name of organic product	butanoate ion or sodium butanoate

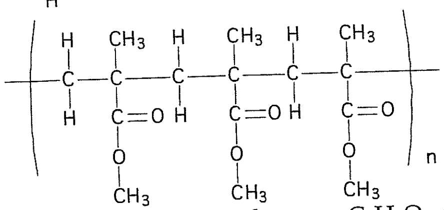
22.(2019:38) a) Note – if asked for two reasons do not give three or more. Only the first two will be marked.

Use	Choice of polymer	Justification
Skylight	polymethyl methacrylate	• moderate UV resistance is more desirable than low UV resistance as it serves to provide protection from harmful UV radiation • lightweight is more desirable as it minimises added weight to the roof structure
Safety glasses	polycarbonate	• in order to protect eyes, it must have at least moderate chemical resistance to withstand chemical splashes, the low chemical resistance of polymethacrylate would be insufficient. • higher heat resistance is required to avoid melting and burning the wearer when protection from heat or sparks is required.

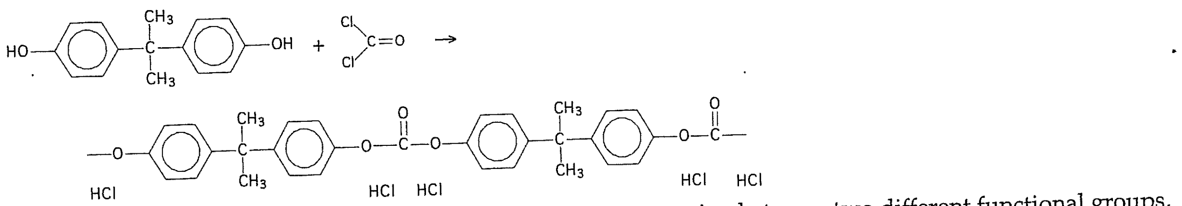
b)



c)

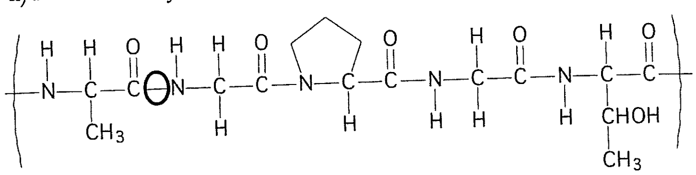


d) Assume a 1:1 mol ratio between C_4H_8O and $C_4H_6O_2$
$n(C_4H_6O_2) = 1.50 \times 10^6/86.088 = 1.742 \times 10^4$ mol
$m(C_4H_8O) = 72.104 \times 1.742 \times 10^4 = 1.256 \times 10^6$ g represents 65% efficiency
$\therefore$ 100% efficiency would be $= 100/65 \times 1.256 \times 10^6 = 1.93 \times 10^6$ g
e) Polymethyl methacrylate is produced from one type of monomer, methyl methacrylate (see (*2019:38c*)). The monomers contain a carbon-carbon double bond across which two species are added.

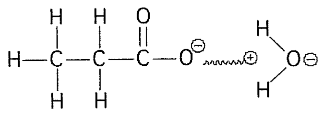


Polycarbonate is produced as shown above. A condensation reaction between two different functional groups, producing a polymer and 'a small molecule' – in this case hydrogen chloride. Remember condensation reactions do not always have to produce water – e.g. in other reactions the small molecule can be ammonia or acetic acid.

23.(2019:41) a) Alanine – Glycine – Proline – Glycine – Threonine
b)



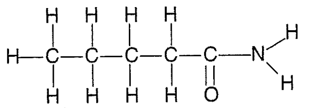
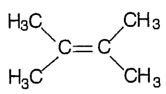
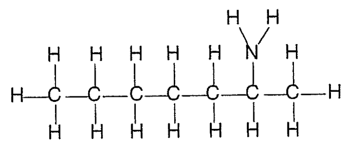
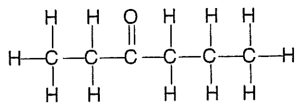
c) The primary structure of a protein is the sequence of alpha amino acids.
The secondary structure is the form taken when amide and carbonyl groups interact to form either alpha helices or beta pleated sheets.
d) Polar carbon compounds interact by dipole–dipole and dispersion forces. They are able to form strong dipole–dipole bonds with water molecules which themselves exhibit the sum of hydrogen bonds, dipole–dipole forces and dispersion forces. The sum of these solute-solvent forces are strong enough to overcome the sum of the solvent–solvent and solute–solute forces allowing the polar carbon compounds to dissolve.

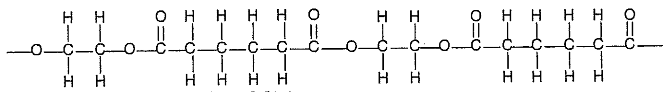


Dipole-dipole force

Protein is held together by strong, covalent, peptide bonds in the primary structure and is folded by the secondary and tertiary structures. It is a compact, large molecule with huge dispersion forces.
The sum of the forces formed with water are not strong enough to overcome either the solute–solute forces nor the solvent-solvent forces making protein insoluble in water.

24.(2020:26)

Full structural formula	**IUPAC name**
	pentanamide
	trans–2,3–dimethylbut–2–ene
	heptan–2–amine
	hexan–3–one

25.(2020:28) a) 

b) i) Polyester ii) Condensation c) Addition

26.(2020:33) This question requires knowledge of hydration, oxidation, esterification and catalysts. Condensed structural formulas are recommended as often atoms are miscounted when writing molecular formulae.

<u>Step One</u>: hydration – requires, steam and phosphoric acid catalyst

$C_2H_4(g) + H_2O(g) \rightleftharpoons CH_3CH_2OH(g)$

<u>Step Two</u>: oxidation – requires an alcohol, and an oxidising agent such as acidified $KMnO_4(aq)$. Either pentanal or pentan-1-ol can be used and a fully balanced equation shown.

$5\ CH_3CH_2CH_2CH_2CH_2OH(aq) + 4\ MnO_4^-(aq) + 12\ H^+(aq) \rightarrow 5\ CH_3CH_2CH_2CH_2COOH(aq) + 4\ Mn^{2+}(aq) + 11\ H_2O(\ell)$

<u>Step Three</u>: esterification – with a sulfuric acid catalyst.

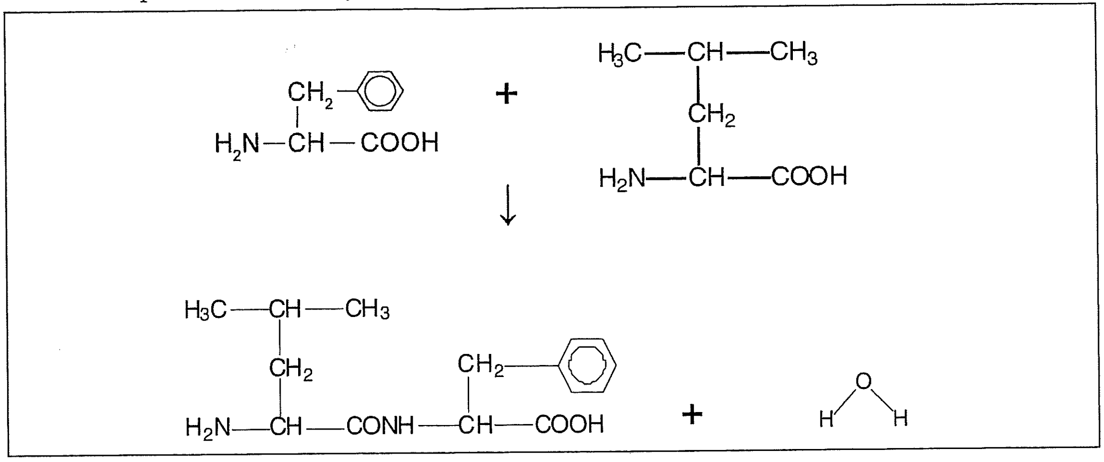
$CH_3CH_2OH(\ell) + CH_3CH_2CH_2CH_2COOH(\ell) \rightleftharpoons CH_3CH_2CH_2CH_2COOCH_2CH_3(\ell) + H_2O(\ell)$

27.(2020:35) a) A definition question – Primary structure – the covalently bonded amino acid chain.
b) We should refer to the Data booklet for the identity of each amino acid.
There are five α-amino acids that are common to both types of Cytochrome C. They both contain lysine.
The only difference is that in position three where human Cytochrome C has serine, grey whale Cytochrome C has alanine.
c)

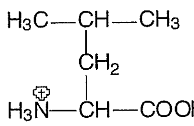
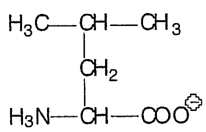
α-amino acid pairs	**Predominant side chain interaction**
Ala and Val	dispersion forces
Gin and His	hydrogen bonding
Cys and Cys	disulfide bridge

d) A definition question - Secondary structure.

e)



f)

Structural formula of leucine	pH
	acidic
	basic

Chapter 11: Synthesis

1.(2015:33) a) $CH_3-(CH_2)n-$ ◎ $-SO_3^-Na^+$ or $CH_3-(CH_2)n-SO_3-Na^+$

b) The long chain hydrocarbon tail is non-polar so its main force of attraction is through dispersion forces. This end of the detergent is most soluble in non-polar substances such as oil, fats and grease.

The charged end of the detergent is ionic so the predominant forces of attraction are ion-dipole forces. Thus detergent is most soluble in polar substances such as water.

c) Soaps combine with Ca^{2+} and/or Mg^{2+} ions in hard water to form a precipitate while detergents do not form precipitates

2.(2016 SP:37) a)

	Temperature (°C)	Pressure (kPA)	Raw material
fermentation	60	101.3	plant material
hydration of ethene	300	7000	crude oil

b) The process uses enzymes to catalyse reactions at lower temperatures where the ethanol does not boil off and the enzymes do not die. The reaction procures economic.rates at lower temperatures lower than industrial catalysts

c) The process uses renewable raw materials, has lower emissions, and can be produced from waste material

3.(2016 SP:40) a) lipase

b) n(veg oil) = $1.50 \times 10^6/855.334 = 1.754 \times 10^3$ mol

$n(CH_3OH) = 3 \times$ n(veg oil) $= 5.261 \times 10^3$ mol

$m(CH_3OH) = 5.261 \times 10^3 \times 32.042 = 1.69 \times 10^5$ g

c) for 100% efficient: n(A) = n(Veg oil) = 1.754×10^3 mol

78% efficient, thus n(A) = $0.78 \times 1.754 \times 10^3 = 1.368 \times 10^3$ mol

MF Ester A is $C_{17}H_{34}O_2$ thus M(A) = 270.442 g mol^{-1}

m(A) = $1.368 \times 10^3 \times 270.442 = 3.70 \times 10^5$ g